Computer Science Department

Software Engineering & Business Analysis

**Bachelor's Thesis**

# MovieCrush

## Movie Tracking Platform with Personalized Recommendations, Detailed Ratings, and Social Features

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*Submission date*
16 June 2026

# TODOS

# Academic Integrity Statement

I, undersigned, hereby declare that this capstone project is the result of my own work.

- All ideas, data, figures and text from other authors have been clearly cited and listed in the bibliography.
- No part of this project has been submitted previously for academic credit in this or any other institution.
- All code, diagrams, and third-party materials are either my original work or are used with permission and properly referenced.
- I have not engaged in plagiarism or any form of academic dishonesty.
- Any assistance received (e.g. from peers, tutors, or online forums) is acknowledged in the acknowledgements section.

I understand that failure to comply with these declarations constitutes academic misconduct and may lead to disciplinary action.

| | |
|---|---|
| Place, date | Kyiv, 02.06.2026 |
| Signature | ______________________ |

# Contents

# Acknowledgements

First of all, I want to thank my parents. For giving me the opportunity to study at such a great university, for their support every single time, for always believing in me when I didn't believe in myself - they always knew I would make it. They have always been my biggest role models in life, and everything I have now is because of them. I also want to thank my mom for being someone I can tell absolutely everything to - every secret, every worry, every thought. And I want to thank my dad for teaching me tennis when I was a child - that's what led me to become the head of the KSE Table Tennis Club and find amazing friends at university, who made these student years the best ones.

I want to thank my grandparents - they were always ready to help me with anything, and were even willing to learn how to code just to make things easier for me. Thank you for the fact that when I had classes until 9 PM, I could get to Akademmistechko in 20 minutes instead of an hour and a half, and thank you for always cooking so much and so well every time I came over, as if it was a wedding and not just my last class of the day. I want to thank my brother for always showing up - early or late - even though he wasn't a student here at that moment, just to help or support me.

I want to thank Veronika Anepska for absolutely everything - for always listening to me, for every party, every conversation, every idea, for believing in each other and supporting each other. Together we built discipline - not just opening TikTok every day to keep our streak alive, but also learning Italian every day, doing yoga every day, setting goals and reaching them together. I'm really happy that this diploma was made with you. I'm sure we have an amazing future ahead, because we're doing everything to make it happen.

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Thank you to Ivan Suprun - after talking to you, everything always feels lighter. Thank you to Marko Hlibovytskyi for always telling me I would be fine when I felt like giving up, and for helping me install Windows when I needed it.

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I also want to thank Kateryna Pyvovarova for our conversation on the 5th floor of KSE - it was you who came up with the Soulmate feature idea, and I'm really glad that conversation happened.

Thank you to absolutely everyone who came into my life during these years and made it the best it could be - in such difficult times.

# Abstract

*The abstract serves as a concise summary of your entire thesis, encapsulating key elements on a single page such as:*
- *General background information*
- *Objective(s)*
- *Approach and method*
- *Conclusions*

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# 1 Introduction

*Your introduction serves to introduce the topic of your Bachelor thesis and to arouse the reader's curiosity with an overview. Why it is important and how it is structured, we explain here.*

*You can consider an introduction as a teaser for your bachelor thesis. You arouse interest and give a foretaste by presenting your motivation, your method and the state of research in your introduction.*

*Convince your examiners already in the introduction that your Bachelor thesis will be exciting. If your professor starts reading your thesis with anticipation and interest, the chances of getting good grades are higher.*

*Pay particular attention to the following in your introduction:*

- ***Introduce the topic*** *- What characterizes the topic?*
- ***Introduce the goal*** *- What do you want to achieve with your thesis?*
- ***Make the reader curious*** *- What motivates the reader to read on?*
- ***Describe the relevance*** *- Why is this bachelor thesis scientifically relevant?*

*The introduction should have the following content:*

- ***Initial situation presentation of the topic*** *- You introduce the topic with an exciting 'bait'. You provide initial information on the topic and the object of research and explain the current state of research.*
- ***Relevance of the topic motivation*** *- You justify the relevance of your topic (scientifically) and place it in the context of your field. In addition, it is often required that you disclose your personal motivation.*
- ***Objectives*** *- Your introduction should clearly state what the goal of your paper is and what outcome you hope to achieve upon completion of the bachelor thesis.*
- ***Method*** *- You explain the approach and justify the choice of method.*
- ***Structure of the Bachelor's thesis*** *- Finally, you give the reader a general overview of your Bachelor's thesis by explaining the structure, showing the red thread and how the research question is answered.*

> Welcome to the template's introductory chapter! Instead of boring you with lorem ipsum, here's a quick guide to what you can do in Typst and, more specifically, in this template.
>
> Need more? Check out the Guide to Typst.

## 1.1 Basic markup

Typst lets you create bold, italic, or monospaced text with ease. You can also sprinkle in equations like $e^{i\pi} + 1 = 0$ or even inline code like `fn main() { println!("Hello, World!") }`. And because life is better in color: pink, blue, yellow, orange, green, and more! **Boldly colorize!**

You can also write numbered or unnumbered lists:

- First item
- Second item
  1. First Subitem

   2. Second Subitem
- Third item

Need equations? Sure! They look great as blocks too:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

## 1.2 Images

As they say, a picture is worth a thousand words. Let's add one:

Figure 2: Project logo

## 1.3 Tables

Tables are great for organizing data. From simple to complex, Typst handles them all:

| **Name** | **Age** | **City** |
|---|---|---|
| Albert Einstein | 25 | Bern |
| Marie Curie | 22 | Paris |
| Isaac Newton | 30 | London |

Table 1: Simple table

| [31:27] | | | [24:20] | [19:15] | [14:12] | [11:7] | [6:0] |
|---|---|---|---|---|---|---|---|
| funct5 | aq | rl | rs2 | rs1 | funct3 | rd | opcode |
| 5 | | | 5 | 5 | 3 | 5 | 7 |

Table 2: Complex table

## 1.4 Boxes

Highlight key points with these fun boxes (and more):

Infobox: For highlighting information.

Ideabox: Share a brilliant idea.

Warningbox: Proceed with caution!

Firebox: This is 🔥 !

Rocketbox: Shoot for the moon!

Todobox: Just do it!

TODO

Personnal todo before marking this thesis as final

## 1.5 Citations, Acronyms and Glossary

Add citations with `@` like [1] or [1, p.7ff] (stored in **`/tail/bibliography.bib`**).

Acronym terms like Infotronics (IT) expand on first use and abbreviate after IT. Glossary items such as Rust Programming Language (Rust) can also be used to show their description as such: Rust is a modern systems programming language focused on safety, speed, and concurrency. It prevents common programming errors such as null pointer dereferencing and data races at compile time, making it a preferred choice for performance-critical applications.. Acronyms and glossary entries auto-generate at the document's end (defined in **`/tail/glossary.typ`**).

## 1.6 Code

Besides writing inline code as such `fn main() { println!("Hello World") }` you can also write code blocks like this:

```rust
fn main() {
  let ship = Starship::new("USS Rustacean", (0.0, 0.0, 0.0));
  let destination = (42.0, 13.0, 7.0);
  let warp = ship.optimal_warp(ship.distance_to(destination));

  println!("🖐 {} traveling to {:?} at Warp {:.2}", ship.name, destination, warp);
  if warp <= 9.0 {
    println!("✨ Warp engaged!");
  } else {
    println!("⚠ Warp failed!");
  }
}
```

Listing 1: First part of the USS-Rustacean code

or directly from a file

```rust
struct Starship {
  name: String,
  position: (f64, f64, f64),
}

impl Starship {
  fn new(name: &str, position: (f64, f64, f64)) -> Self {
    Self {
      name: name.into(),
      position,
    }
  }
  fn distance_to(&self, dest: (f64, f64, f64)) -> f64 {
    ((dest.0 - self.position.0).powi(2)
      + (dest.1 - self.position.1).powi(2)
      + (dest.2 - self.position.2).powi(2))
    .sqrt()
  }
  fn optimal_warp(&self, distance: f64) -> f64 {
    (distance / 10.0).sqrt().min(9.0)
  }
}
```

Listing 2: Second part of the USS-Rustacean code from `/resources/code/uss-rustacean.rs`

## 1.7 Context Problem

Haute École d'Ingénierie (HEI) Rust Rust programs

[1], [1, p.7ff]

```rust
fn main() {
  println!("Hello World!");
}
```

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

## 1.8 Objectives

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## 1.9 Structure of this report

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

# 2 Research

## 2.1 Research Methodology

To validate the problem and understand user needs before building MovieCrush, three complementary research methods were used:

- **Quantitative survey** - a structured questionnaire distributed to 262 respondents to measure user behavior, pain points, and feature demand at scale.
- **Qualitative user interviews** - five in-depth interviews conducted to uncover motivations, frustrations, and unmet needs that surveys cannot capture.
- **Competitor analysis** - a systematic review of existing film tracking platforms to identify gaps in the market and opportunities for differentiation.

All three methods were applied before development began, ensuring that product decisions were driven by real user needs and market evidence rather than assumptions.

## 2.2 Business Process Research

### 2.2.1 Stakeholder Analysis

Understanding who is affected by MovieCrush and how they interact with the product is essential for making the right design decisions. The table below outlines the key stakeholders, their interests, influence on the project, and the engagement strategy.

| Stakeholder | Interest | Influence | Risk / Opportunity | Strategy |
|---|---|---|---|---|
| End users (film enthusiasts) | Discover content, track watches, find like-minded people | HIGH. They define product success | Risk: low loyalty, low premium conversion | Deep user research, fast iteration based on feedback |
| Streaming services (Netflix, MEGOGO) | Grow and retain subscribers | MEDIUM. May or may not provide affiliate opportunities | Opportunity: affiliate revenue. Risk: refusal to cooperate | Position MovieCrush as a traffic source for their platforms |
| Data providers (TMDb API) | Monetize their database, grow API usage | HIGH. Without their data the app is impossible | Risk: API terms change, price increases, quota limits | Use free tier at launch, plan migration to paid tier at scale |
| AI providers (Google Gemini) | Monetize AI models | HIGH. Recommendation quality defines product uniqueness | Risk: high API costs, unpredictable policy changes | Optimize requests, budget for AI costs from the start |
| Competitors (Letterboxd, TV Time) | Protect market share | MEDIUM. May copy features or run aggressive marketing | Risk: user churn | Focus on unique features (Soulmate, Wrapped) |

Table 3: MovieCrush stakeholder analysis

## 2.3 Product Discovery and Business Analysis

Before writing any code, a business analysis was done to define what the product should be, who it is for, and how users would move through it. The full discovery board is available at: Miro Board - MovieCrush Business Analysis.

The board shows the original product vision from before development started. Some features changed or were removed during the process based on what was realistic to build - this is mentioned throughout the thesis where it matters.

### 2.3.1 Vision and Mission

**Vision:** To create a home for movie lovers - a place to share impressions, discover new films through friends, and relive every emotion. MovieCrush helps you find not just your next favorite film, but also the people who will love it with you.

**Mission:** To solve the "what should I watch?" problem for good. Using personalization and community to help users discover, track, and share movies they will truly love - making every viewing decision easy and every opinion heard.

### 2.3.2 Value Proposition

The Value Proposition Canvas below shows the main user problems, what users are trying to do, and what MovieCrush offers in response.

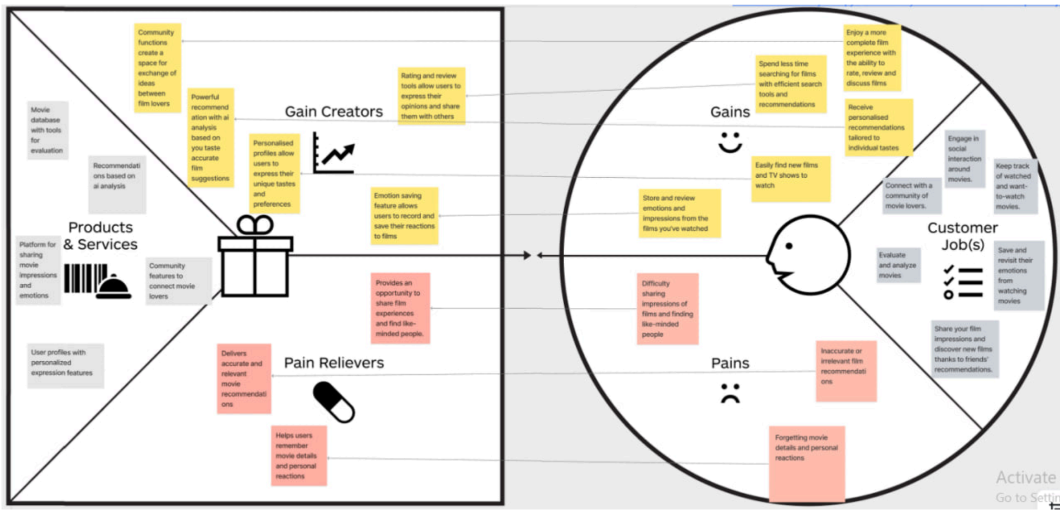


Figure 3: Value Proposition Canvas developed during the discovery phase

### 2.3.3 Customer Journey

The customer journey map covers five stages: Awareness, Consideration, Purchase, Engagement, and Renewal. It shows how users find out about MovieCrush, what makes them sign up, and what keeps them using it.

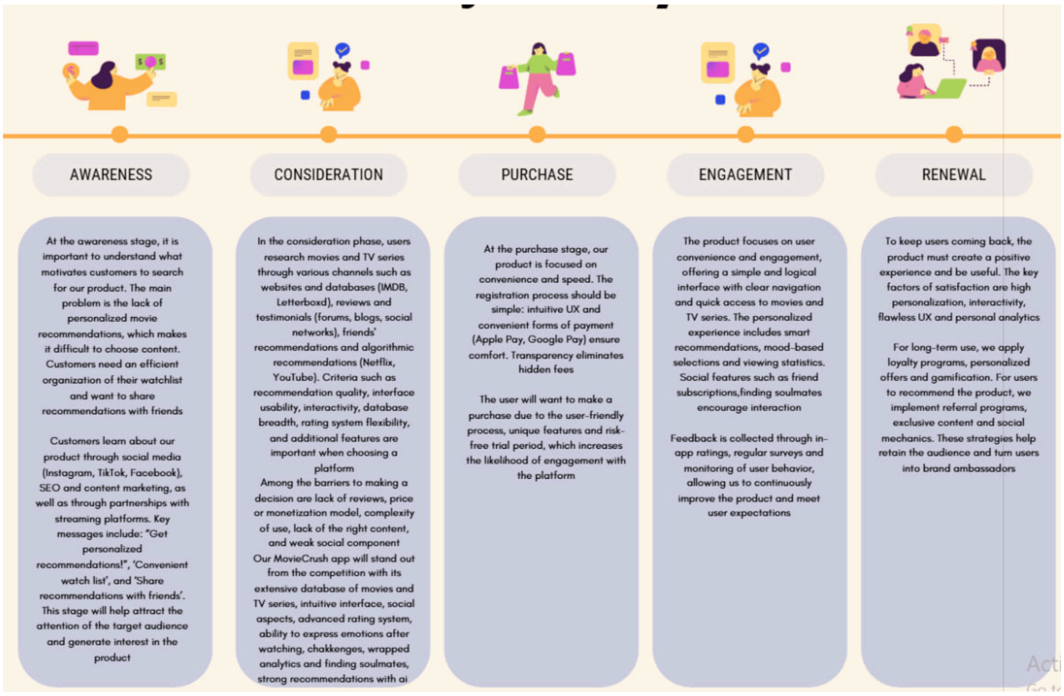


Figure 4: Customer Journey Map - from discovery to long-term retention

## 2.4 Quantitative Research: User Survey

The survey consisted of 5 blocks of questions:

- **Demographics:** age, content language, profession
- **Digital habits:** streaming subscriptions, monthly spend
- **Viewing habits & pain points:** how many films per week, how they currently track films, what they like/dislike about existing apps
- **Feature validation:** interest in MovieCrush features (AI recommendations, Wrapped, Soulmate, detailed ratings, challenges)
- **Monetization:** attitude to ads, willingness to pay for premium (3 USD/month)

The full survey is available here. Raw responses from all 262 respondents are available in Google Sheets.

### 2.4.1 Survey Distribution & Incentive

To maximize participation, the survey was distributed via Slack to the KSE student community. To motivate respondents, 4 cinema tickets were raffled among participants - 2 winners, 2 tickets each for any film and any date of their choice.

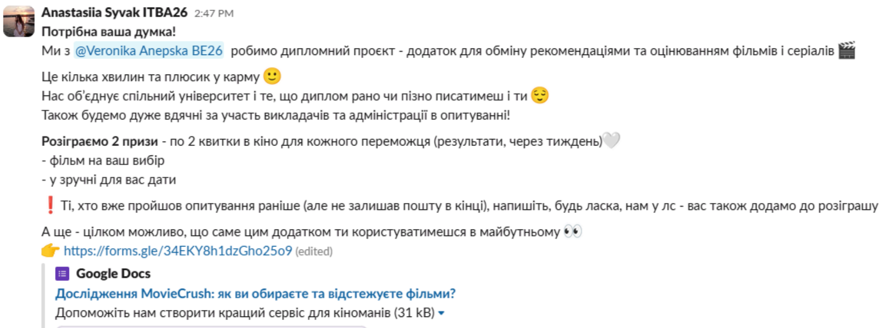


Figure 5: Survey announcement in KSE Slack

The announcement generated strong engagement - 43 reactions, which is significantly higher than typical KSE Slack posts that receive 2-6.

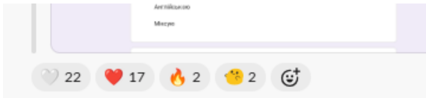


Figure 6: Community reactions - 43 emoji responses to the announcement

Both winners were contacted and confirmed receipt of their cinema tickets.

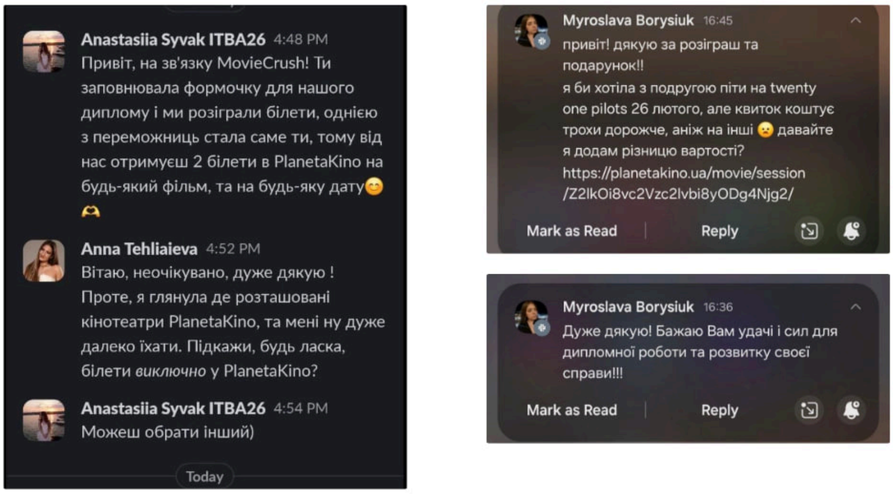


Figure 7: Winner confirmation messages

### 2.4.2 Demographics

The survey collected 262 responses. The majority of respondents - **90.5%** - were aged 18-24, which reflects the author's social circle and the primary target audience for MovieCrush. **82.7%** have active streaming subscriptions, confirming high purchasing power and an established habit of paying for digital content.

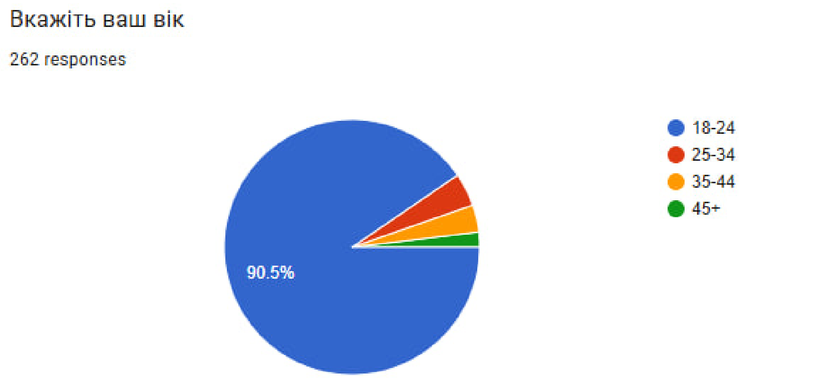


Figure 8: Age distribution of survey respondents

In terms of content language, **42.4%** consume content in a mix of Ukrainian and English, **36.3%** primarily in Ukrainian, and **21.4%** in English only - confirming that the app must support both languages. Some respondents mentioned russian-language content. The author's position is clear: russian will not be supported. MovieCrush targets Ukraine, Europe, and the US - not the CIS market.


Figure 9: Language in which respondents consume film content

### 2.4.3 Viewing Habits & Pain Points

**43.1%** of respondents watch 2-5 films or series per week - the core active audience that needs tools for tracking and recommendations.


Figure 10: Average number of films or series watched per week

When asked how they currently track films, **49.8%** said they use phone notes - a clear signal that no proper tool exists. Only 19.8% use dedicated services like Letterboxd or IMDb, suggesting low awareness or poor fit with existing solutions. Additionally, 7.6% use TikTok or Instagram to save films - a new behavioral pattern no existing app addresses. 24.9% do not track films at all, representing untapped latent demand.

| Як саме ви зараз запам'ятовуєте/відстежуєте фільми, які хочете подивитися або вже подивилися? | | |
|---|---|---|
| Не відстежую | 59 | 24.9% |
| Нотатки в телефоні | 118 | 49.8% |
| Використання наших конкурентів | 47 | 19.8% |
| Збережені в тік ток/інстаграм | 18 | 7.6% |
| Списки в стрімінгах (Netflix) | 39 | 16.5% |

Figure 11: How respondents currently track films they want to watch or have watched

### 2.4.4 Competitor Pain Points

When asked what bothers them about existing services like IMDb and Letterboxd, respondents highlighted recurring problems:

- **Poor recommendation algorithms** - platforms suggest popular content, not personalized picks
- **No TV series or anime** - Letterboxd ignores a huge segment of content
- **No statistics** - users want a Spotify Wrapped equivalent for films
- **Outdated interface** - IMDb is losing younger users due to poor design
- **No Ukrainian language** - a barrier for the mass Ukrainian audience
- **Fragmented lists** - content saved across multiple platforms with no unified view

At the same time, users value what currently works: trust in IMDb ratings, seeing what friends watch, viewing diaries, and watchlists. These became the foundation for MovieCrush's core feature set.

### 2.4.5 Where Users Look for Recommendations

**83.6%** of respondents look for recommendations on social media (Instagram, TikTok, Twitter), and **76%** rely on friends. Only **7.3%** use specialized apps like Letterboxd or TV Time. This confirms that existing platforms are failing at discovery.

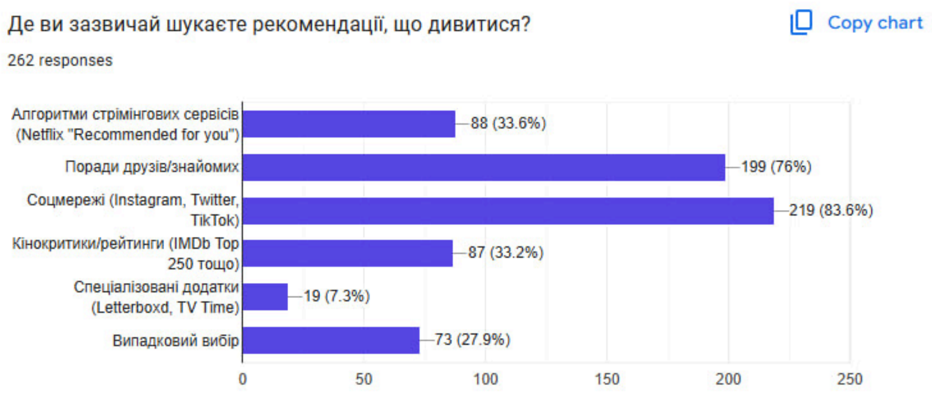


Figure 12: Where respondents usually look for recommendations on what to watch

**58%** actively discuss films with friends and family, and **37%** do so occasionally - confirming strong demand for social features.

### 2.4.6 Feature Validation

Respondents were asked which features they would be willing to pay for. The top two were the **hybrid recommendation system** (42%) and **personal statistics/Wrapped** (35.9%). Social features like Soulmate (14.5%) showed lower willingness to pay but high viral potential - making them a strong fit for the free tier.


Figure 13: Features respondents would be willing to pay for separately

It is worth noting that during the survey, the planned feature was described as an "AI assistant." In the final implementation, this became a hybrid recommendation system combining ALS collaborative filtering with TMDB Discover for candidate generation, and Gemini for re-ranking - a more technically sound solution to the same user need.

The personal statistics feature (Wrapped) received an average rating of **4.24 out of 5** - the highest of all features tested. Full rating distribution is available in the Appendix.

### 2.4.7 Social Features

When asked which social feature mattered most, **48.9%** chose seeing what their friends are watching and rating. **38.5%** wanted shared lists, and **35.1%** wanted to compare tastes with friends. This strongly validates the social layer of MovieCrush.


Figure 14: Most important social features according to respondents

### 2.4.8 Monetization Insights

**67.6%** of respondents would accept ads in a free version, validating the freemium model. **23.3%** said ads would bother them even in a free version - this segment is the guaranteed base for premium conversion.

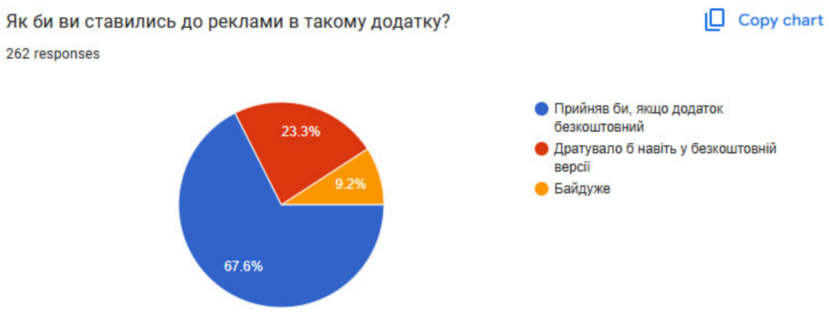


Figure 15: Respondent attitude toward ads in the app

On pricing, **75.5%** of respondents would consider paying 3 USD or more per month for a premium package including AI recommendations, extended statistics, Soulmate search, and no ads. Only **9.9%** said they would not pay at all.

Figure 16: Maximum monthly price respondents would consider for a premium subscription

### 2.4.9 Open-ended Responses

The final open-ended question asked what would motivate users to switch to a new film app. The most frequent themes across 262 responses were:

| Theme | Mentions |
|---|---|
| AI / Personalized recommendations | 25+ |
| Clean and beautiful interface | 20+ |
| Statistics / Wrapped | 15+ |
| Social features (friends, lists) | 15+ |
| Large library / anime / series | 12+ |
| Support for Ukrainian product | 10+ |
| Unique features (Soulmate, challenges) | 8+ |
| Price / free access | 7+ |
| Niche content | 5+ |

Table 4: Most frequent themes in open-ended responses

### 2.4.10 Ideal Customer Profile

Based on all survey findings, the target segment is:

- **Age:** 18-24 years old (90.5% of respondents)
- **Status:** Students (32.3%) and young IT professionals (17.7%)
- **Language:** Mix of Ukrainian and English (42.4%)
- **Viewing habits:** 2-5 films/series per week (43.1%)
- **Spending:** up to 25 USD/month on subscriptions
- **Pain point:** Using phone notes to track films (49.8%)
- **Wants:** hybrid AI recommendations (42%) and personal statistics (35.9%)

> **Strong Product-Segment Fit** for: "Ukrainian tech-oriented youth aged 18-24, actively consuming content, unsatisfied with existing solutions, and willing to pay 3 USD/month for a quality personalized experience."

## 2.5 Qualitative Research: User Interviews

Five in-depth interviews were conducted with target users before development began. Recordings are available at: Telegram channel.

### 2.5.1 Methodology

In-depth interviews were chosen over surveys because open-ended questions force respondents to formulate their own opinions rather than selecting from predefined options. Individual format allows the interviewer to follow unexpected threads and uncover insights that are not on the surface. Group interviews were avoided to prevent social influence on answers.

Respondents were selected to represent different life situations: two students, one working professional, and two family respondents. The only selection criterion was an interest in watching films or series. Unlike the quantitative survey - where 90.5% of respondents were aged 18-24 - the interview participants represented a broader age range. This was intentional: to capture different behavioral patterns and avoid confirming only the assumptions of one demographic group.

Each interview was conducted at a convenient time and location for the respondent. All participants gave verbal consent to recording and further use of the data.

### 2.5.2 Interview Findings

**Time spent searching varies dramatically.** Some respondents find something in 5 minutes, others spend up to 3 hours. Regardless of the actual time, all respondents considered the search process cognitively exhausting:

> "If within 2-3 minutes I can't find anything - the desire just passes and I don't watch anything at all."

**Phone notes are universally used but universally disliked.** Every respondent who keeps lists uses phone notes or saved TikToks. All of them described this as inconvenient:

> "It's not convenient because there are too many actions - you always need to adjust it, add things, delete things."

**TikTok and Instagram edits are the primary discovery tool.** Respondents do not rely on streaming algorithms to discover new content. Instead, they use short video edits to decide whether a film matches their mood before committing to watch it.

**Friends' recommendations are trusted more than ratings.** IMDb ratings were specifically criticized for being misleading:

> "I don't like IMDb ratings - they often spoil my impression of a film. I would rate it 10, but IMDb shows 3."

**Filters are the most requested feature.** When asked what they want in an app, almost every respondent mentioned filters:

> "The more filters, the better - the faster you can find what you want."

**A successful recommendation creates a strong emotional reaction.** When asked how they feel after finding the perfect film, respondents used words like "euphoria" and "delight" - confirming that the emotional payoff of a good recommendation is significant.

### 2.5.3 User Persona

Based on the interviews, the following persona was developed:

| | |
|---|---|
| **Name** | Anna |
| **Age** | 25 |
| **Status** | Junior specialist, active schedule |
| **Platforms** | Netflix, HD Rezka, YouTube |
| **Discovery** | TikTok edits, Instagram, friends |
| **Tracking** | Phone notes - finds it inconvenient |
| **Pain point** | Spends too much time searching, recommendations don't match expectations |
| **Motivation** | Emotional relief after a busy week |
| **Key request** | Better filters, see what friends watch |

Table 5: User persona derived from qualitative interviews

### 2.5.4 Implications for MovieCrush

The interviews directly shaped the following product decisions:

- **Mood-based filters** - respondents choose films by mood, not genre alone
- **Social feed** - seeing what friends watch is more trusted than any algorithm
- **Detailed ratings with comments** - numeric ratings are not trusted without context
- **Simple onboarding** - "the more actions required, the less likely people are to use it"

- **Watchlist and tracking** - replacing phone notes with a proper tool was a universal need

## 2.6 Competitor Analysis

### 2.6.1 Overview of Existing Solutions

The competitive landscape for film tracking apps includes four main players, each with a distinct approach and clear limitations that MovieCrush addresses.

#### 2.6.1.1 Letterboxd

Full analysis with screenshots

Founded in 2011 in New Zealand, Letterboxd is the market leader in social film tracking with 17 million registered users as of 2024. Its strengths are a strong community, detailed film pages, and a well-known brand among film enthusiasts. The main weaknesses are that it covers films only, no TV series, has no personalized recommendations, and locks statistics behind a paid subscription. Its annual revenue is estimated at 5.5-19 million USD.

#### 2.6.1.2 TV Time

Full analysis with screenshots

Founded in 2013 in France, TV Time has around 30 million registered users and focuses on TV series tracking. It is free for users but monetizes through selling anonymized viewing data to studios and streaming services - a model that creates a conflict of interest between the product and its users. Key weaknesses include a poor interface, no friend ratings visible, weak recommendations, and problems with Ukrainian language support.

#### 2.6.1.3 IMDb

Full analysis with screenshots

Founded in 1990 and owned by Amazon, IMDb is the world's largest film and series database with over 670 million monthly site visits. It is a reference tool rather than a social platform - there are no friend connections or communities. Registration requires an Amazon account, the interface is overloaded, and the platform is deeply tied to the Amazon ecosystem which limits its independence as a product.

#### 2.6.1.4 Moviebase

Full analysis with screenshots

A German indie project launched in 2017, Moviebase uses TMDB as its data source and supports 39 languages. It has good tracking features and an anti-spoiler system in comments, but has no social features at all, no gamification, and is only available on Android. 53% of its downloads come from Brazil, which limits its appeal in other markets. Annual revenue is estimated at 50,000-60,000 USD.

#### 2.6.1.5 Market Comparison

| Metric | Letterboxd | TV Time | IMDb | Moviebase |
|---|---|---|---|---|
| Registered users | 17M | 30M | 100M+ | 500K |
| Content | Films only | Series + films | Films + series | Films + series |
| Social features | Basic | Groups only | None | None |
| Personalized recs | No | Poor | Basic | Basic |
| Statistics / Wrapped | Pro only | No | No | Pro only |
| Soulmate / matching | No | No | No | No |
| Ukrainian language | No | Partial | Yes | Yes |
| Revenue model | Subscriptions + ads | Data sales | Ads + Amazon | Subscriptions + ads |
| Est. annual revenue | 5.5-19M USD | 4.2M USD | 224M USD | 50-60K USD |

Table 6: Key metrics comparison across competing platforms

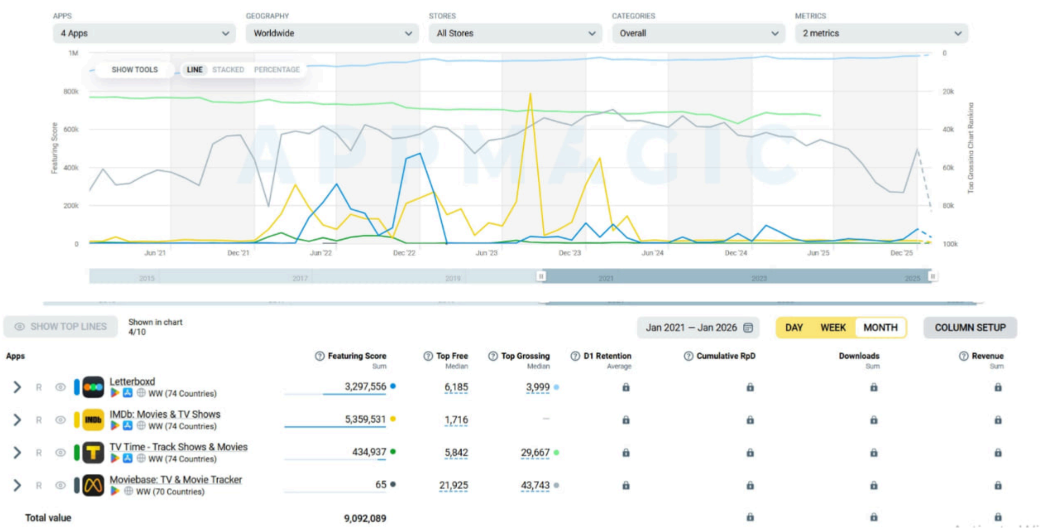


Figure 17: Featuring Score and chart rankings across four competitors, Jan 2021 - Jan 2026 (AppMagic). Letterboxd leads in Featuring Score with 3.3M, while Moviebase scores just 65 - reflecting the large gap in platform visibility and store promotion.

### 2.6.2 Competitor Gap Analysis

The table below summarizes the key weaknesses identified in competing platforms and how MovieCrush addresses each of them.

| Competitor | Key Weaknesses | How MovieCrush Addresses Them |
|---|---|---|
| **Letterboxd** | 1. Films only - no TV series support<br>2. No personalized recommendations<br>3. Limited social interaction (likes and comments only) | 1. Unified catalog: films, series, and individual episodes<br>2. Hybrid recommendation system: ALS collaborative filtering with Gemini re-ranking<br>3. Deep social layer: Soulmate, detailed ratings with comments, Best Actor voting |
| **TV Time** | 1. Poor UX and outdated interface<br>2. Weak language support<br>3. Cast-only - no directors or screenwriters<br>4. Poor recommendations<br>5. No direct user monetization (revenue from data only)<br>6. Friends' ratings not visible | 1. Modern, intuitive UI/UX built as a core priority<br>2. Full Ukrainian language support<br>3. Complete credits: actors, directors, screenwriters<br>4. Personalized recommendations via ALS + Gemini<br>5. Direct user monetization through premium subscription<br>6. Social feed showing friends' ratings and activity |
| **IMDb** | 1. No social features (no friends, no communities)<br>2. Overloaded, cluttered interface<br>3. Tied to Amazon ecosystem<br>4. No personal analytics beyond basic counters | 1. Social layer as a core product feature: follows, Soulmate<br>2. Clean, user-experience-focused design<br>3. Independent service with email-based registration<br>4. Deep personal analytics via Wrapped |
| **Moviebase** | 1. No social features<br>2. No gamification<br>3. Analytics locked behind paywall<br>4. Narrow geographic reach (53% Brazilian audience) | 1. Soulmate and social ratings as core free features<br>2. Gamification planned for future development<br>3. Wrapped analytics available to all users<br>4. Designed for Ukrainian, European, and US markets |

Table 7: Competitor gap analysis and MovieCrush solutions

**Core thesis:** MovieCrush combines the strengths of existing platforms while eliminating their key weaknesses, and introduces unique social features.

## 2.7 Research Limitations

These results reflect a non-representative sample - 89.9% of respondents were aged 18-24. We found **Product-Segment Fit**, not full Product Market Fit.

The following limitations should be considered when interpreting these results:

1. **Non-representative sample** - 89.9% of respondents were aged 18-24, which correlates with the author's age and social circle.
2. **Narrow demographic** - results reflect the views of students in technical and economic fields only.
3. **Conclusion:** We cannot claim full Product Market Fit for the entire market. Instead, we found **Product-Segment Fit** for the segment of "youth aged 18-24, students, technical specializations." Validation on adjacent segments (25-34, non-IT professions) remains a goal for future work.

## 2.8 Economic Analysis

### 2.8.1 Revenue Model

MovieCrush uses a freemium model with two tiers. The free tier includes full tracking, social features, Wrapped analytics, and basic recommendations - more than most competitors offer for free. Revenue at this level comes from ads and affiliate links to streaming platforms.

The paid tier - MovieCrush Pro - unlocks the Soulmate feature, AI-powered recommendations, and Instagram story templates. These features do not exist in competing apps, which justifies a price above the market baseline.

### 2.8.2 Pricing Strategy

Competitor pricing ranges from 13 USD/year (Moviebase) to 49 USD/year (Letterboxd Patron). Letterboxd Pro at 19 USD/year is the closest benchmark - its main selling point is statistics, which MovieCrush gives away for free.

MovieCrush Pro is priced at **23.88 USD/year (1.99 USD/month)** for the annual plan and **2.99 USD/month** for monthly. The 1 USD/month difference between the two creates a clear reason to choose the annual plan.

| Plan | Price | Per month | Includes |
|---|---|---|---|
| Free | 0 USD | - | Full tracking, social features, Wrapped, ads |
| Pro - annual | 23.88 USD/year | 1.99 USD | No ads, Soulmate, AI recommendations, story templates |
| Pro - monthly | 2.99 USD/month | 2.99 USD | Same as annual |

Table 8: MovieCrush pricing tiers

#### 2.8.3 Cost Breakdown

Current monthly infrastructure costs at the student project stage:

| Service | Plan | Monthly cost |
|---|---|---|
| Render - backend | Free tier (current) | 0 USD |
| Render - CF service | Free tier (current) | 0 USD |
| Render Postgres | Basic | 7 USD |
| Google Gemini API | Free tier (current) | 0 USD |
| Resend | Free tier | 0 USD |
| TMDB API | Free tier | 0 USD |
| EAS Build | Free tier | 0 USD |
| **Total (current)** | | **7 USD/month** |
| **Total (at scale)** | | **23-26 USD/month** |

Table 9: Monthly infrastructure costs

At the current stage, almost all services run on free tiers, bringing the actual monthly cost to just 7 USD - only Render Postgres requires a paid plan. At scale, moving to paid tiers for the backend, CF service, and Gemini API would bring the total to around 23-26 USD/month.

#### 2.8.4 Investment Requirements

No external investment was needed to build and launch MovieCrush. All infrastructure runs on free or low-cost tiers. The only real cost was time. If the product were to grow beyond the current scale, the first investment would go toward moving to paid Render plans and a larger Postgres instance to handle more users.

#### 2.8.5 Financial Projections

Using the realistic scenario of 100,000 MAU (220,000 registered users) and a 5% conversion to Pro:

| Source | Annual estimate |
|---|---|
| Pro subscriptions (11,000 users multiply 23.88 USD) | 262,680 USD |
| Ads (free tier users) | 20,000-40,000 USD |
| Affiliate links | 5,000-15,000 USD |
| **Total** | **287,000-317,000 USD** |

Table 10: Revenue projection at 100,000 MAU

These numbers are projections based on market benchmarks, not real revenue. The product is at an early stage and has not yet been monetized.

### 2.8.6 Future Monetization Plans

The next step in monetization is affiliate partnerships with streaming platforms such as JustWatch, where MovieCrush earns a small commission when a user clicks through to watch a film on a streaming service. This model is already used by Letterboxd and fits naturally into the product since users already look for "where to watch" information.

Selling anonymized user data was considered but rejected - it creates a conflict between the product and its users, which goes against the core idea of MovieCrush as a user-first platform.

## 2.9 Domain-Specific Analysis

### 2.9.1 Value Chain

Understanding who contributes value in the film tracking domain helps clarify MovieCrush's dependencies and strategic position.

| Participant | Role | Impact on MovieCrush |
|---|---|---|
| Studios / Rights holders | Own rights to films | Posters and metadata used via TMDb API - direct usage without official API carries DMCA risk |
| Data aggregators (TMDb) | Collect and maintain metadata: cast, descriptions, posters | Critical dependency. TMDb is the primary data source - open, well-maintained, and best suited for localization |
| Streaming services (Netflix, MEGOGO) | Final content providers | Their libraries define what users can watch. Their own recommendation engines are direct competitors |
| Competitors (Letterboxd, TV Time) | Social film tracking platforms | Set UX standards, monetization models, and user expectations |

Table 11: Value chain analysis for the film tracking domain

### 2.9.2 Technical Challenges

Building a film tracking platform involves several domain-specific technical challenges that shaped architectural decisions in MovieCrush:

- **AI recommendation complexity:** Unlike music, where taste is often genre-based, cinematic taste is **multimodal** - a person may love a specific director, cinematographer, actor, atmosphere, or even sound design rather than a genre. This makes building a recommendation model significantly harder. MovieCrush addresses this with a hybrid approach: ALS collaborative filtering learns from behavioral patterns across users and has priority as the main source of candidates. TMDB Discover serves as an additional candidate source. Google Gemini then re-ranks the combined pool based on the individual user's watch history and ratings - capturing taste signals that a simple genre filter cannot.
- **Cold start problem:** New users have no watch history, making collaborative filtering impossible. MovieCrush solves this with a dedicated onboarding flow: users select favorite actors, then receive a curated set of films which they mark as watched or not watched and rate. The system analyzes content type and genre signals from these ratings to seed initial recommendations via TMDB Discover.
- **TMDb caching for analytics:** MovieCrush maintains a local PostgreSQL cache of TMDb metadata (title, poster, genres, cast, director) that is populated when a user marks a film as watched or adds a rating. This cache is used for Wrapped statistics and recommendation building to avoid excessive TMDb API calls during computation-heavy operations. For regular browsing, the app fetches data directly from TMDb with a short-lived in-memory LRU cache to reduce redundant requests within the same session.

### 2.9.3 Legal & Economic Constraints

- **Copyright on visuals:** Using posters and metadata for informational purposes is generally accepted when sourced through official APIs like TMDb. Direct commercial use without proper licensing carries legal risk. MovieCrush uses only TMDb-provided assets within their terms of service.

# 3 | Design

*In this section you turn your requirements into a concrete engineering blueprint. You'll justify every major architectural choice, visualize structure with C4 diagrams for the first three layers, and map out your runtime topology so that peers can understand—and you can defend—every aspect of your system.*

1. *Clarify how functional and non-functional requirements drive your high-level architecture*
2. *List each architectural decision (for example, "We chose microservices to enable independent scaling and deployment") and explain why it best meets your goals*
3. *Include a C4 Context diagram showing your system in its environment (users, external systems, data sources)*
4. *Include a C4 Container diagram breaking the system into deployable units (APIs, web front end, background workers, databases) and annotate communication styles and protocols*
5. *Include a C4 Component diagram for your core container(s), illustrating key modules, services or libraries and their interactions*
6. *Describe your deployment topology: physical or cloud hosts, network zones, load-balancing, failover and backup strategies*
7. *Summarize your technology stack, mapping each tool or framework back to a specific container or component and noting any trade-offs (performance, community support, learning curve)*
8. *Outline how data flows through the system—including storage models, messaging patterns or API contracts—and note any schema or interface versioning plans*
9. *Address cross-cutting concerns (Security, Logging, Monitoring, Scalability) and show where they sit in your topology*

*By walking through Requirements → Decisions → Diagrams → Topology, your Design section becomes a rigorous, evidence-backed foundation for the implementation that follows.*

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## Contents

## 3.1 Section 1

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## 3.2 Section 2

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## 3.3 Conclusion

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# 4 Implementation

*In this section you translate your component-level designs into working code and systems. Focus on the C4 Component layer and on the details needed to show how your design was realized. Include only the most important code snippets that illustrate key patterns or algorithms, rather than full listings.*

1. *Describe the development methodology (for example, Agile or test-driven development) used to guide your implementation*
2. *Explain any prototyping or iterative strategies you applied to refine components before full-scale coding*
3. *Summarize coding standards, naming conventions and architectural patterns followed in your codebase*
4. *Present critical code snippets or configuration templates that highlight how core components were implemented (for example, key classes, interfaces or algorithms)*
5. *Detail your testing approach and quality assurance measures (unit tests, integration tests, coverage metrics)*
6. *Note any performance optimizations or profiling results for components that were bottlenecks*
7. *Outline your deployment and configuration management process for component artifacts (containerization, CI/CD pipelines)*
8. *Highlight documentation deliverables (API references, inline comments, architecture decision records) that support future maintenance*

*This section demonstrates how each component specification becomes actual, maintainable code—closing the loop from design to implementation.*

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## Contents

## 4.1 Section 1

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## 4.2 Section 2

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## 4.3 Conclusion

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# 5 Validation

*Validation (Requirements Verification and Testing)*

*In this section you demonstrate how your implementation satisfies the initial requirements through clear testing methods and concise examples—suitable for a bachelor-level project:*

1. *Restate each key functional and non-functional requirement from your Analysis and Design sections*
2. *Describe the testing approach for each requirement (for example, unit tests, manual acceptance checks or scenario walkthroughs)*
3. *Provide concrete test cases or usage examples that show how you verify each requirement in practice*
4. *Summarize actual versus expected outcomes, indicating pass/fail status for each test*
5. *Include brief snippets of test code or sample console outputs to illustrate your procedures*
6. *Note any gaps or deviations and suggest simple fixes or areas for future improvement*
7. *If a feature wasn't intended for specific scenarios (e.g. high-load), omit unrealistic stress tests and clearly document its current limitations*

*This focused structure ties every requirement directly to validation results, using examples and methods you can realistically carry out at the bachelor level.*

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## Contents

## 5.1 Section 1

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## 5.2 Section 2

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## 5.3 Conclusion

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# 6 Conclusion

*Conclusion*

*In this final section you bring together your work and reflect on its impact. Keep it concise, restating key points without introducing new information:*

1. *Project Summary: Briefly recap objectives, methodology and principal results*
2. *Alignment with Objectives: Discuss how outcomes meet initial goals, referencing requirements and design aims*
3. *Lessons Learned and Challenges: Note any obstacles and how they informed improvements*
4. *Limitations: Acknowledge features or scenarios beyond this scope and clearly state current system boundaries*
5. *Future Work: Suggest practical enhancements or research directions building on your findings*

*Avoid introducing new concepts here; refer readers to the Discussion for deeper analysis.*

## 6.1 Project summary

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## 6.2 Comparison with the initial objectives

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## 6.3 Encountered difficulties

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## 6.4 Future perspectives

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# Glossary

## Programming Language

***Rust*** **– Rust Programming Language**: Rust is a modern systems programming language focused on safety, speed, and concurrency. It prevents common programming errors such as null pointer dereferencing and data races at compile time, making it a preferred choice for performance-critical applications. 6, 7

## University

***HEI*** **– Haute École d'Ingénierie** 7

***IT*** **– Infotronics** 6

# Bibliography

[1] S. Zahno *et al.*, “Dynamic Project Planning with Digital Twin,” *Frontiers in Manufacturing Technology*, vol. 3, May 2023, doi: 10.3389/fmtec.2023.1009633.

# A Appendix

## AA Survey Materials

### AAA Raffle Process

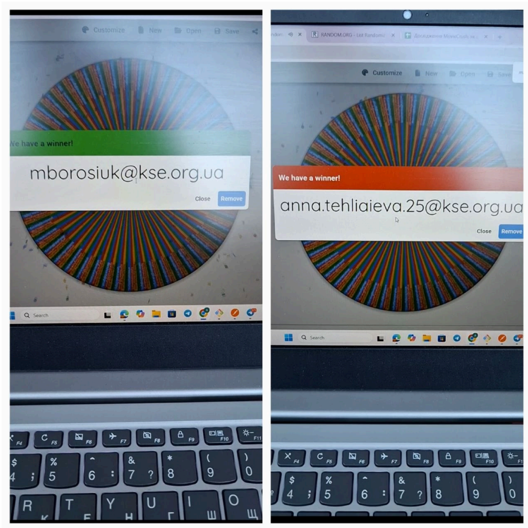


Figure 18: Random wheel spinner used to select raffle winners

### AAB Participant Messages

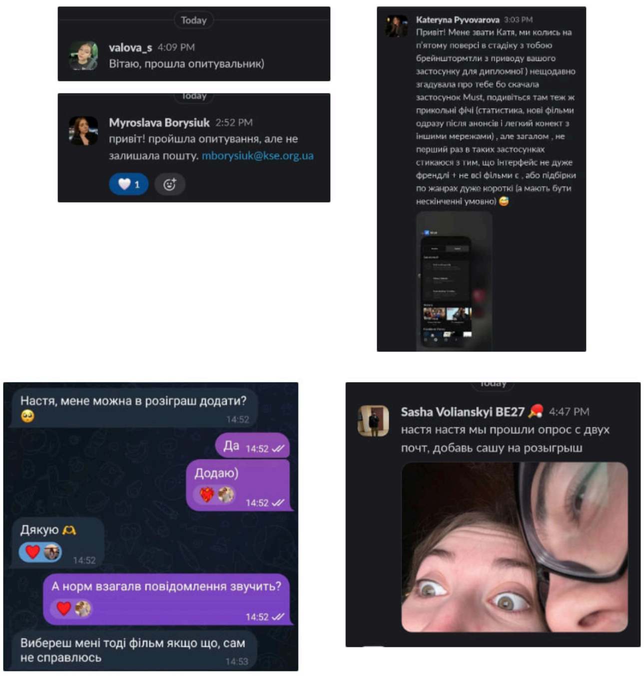


Figure 19: Messages from participants requesting to join the raffle

# AB Additional Survey Charts

*Note: the tables below are based on 237 responses (an earlier snapshot of the survey). The final response count was 262.*

## ABA Wrapped Statistics Feature Rating

The Wrapped feature received the highest average rating of all tested features - **4.24 out of 5** - making it the most anticipated feature among respondents. 53.8% gave it the maximum score of 5.

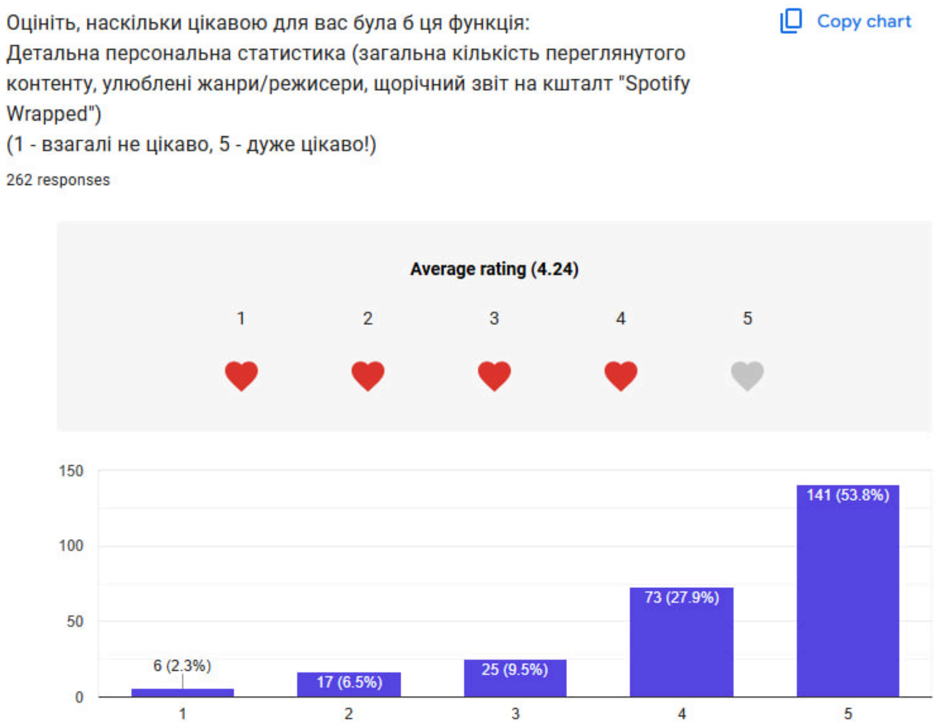


Figure 20: Rating distribution for the Wrapped statistics feature — average 4.24/5

## ABB Monthly Spending on Subscriptions

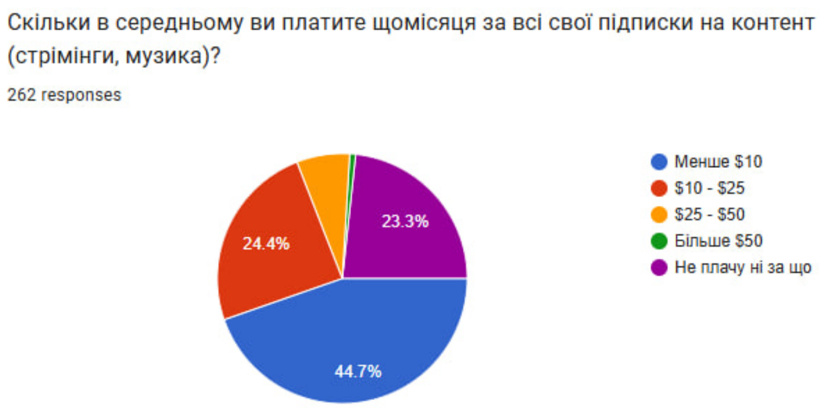


Figure 21: Monthly spending on content subscriptions among respondents

## ABC Active Streaming Subscriptions


Figure 22: Active streaming subscriptions among respondents

| Підписки на стрімінгових платформах | Кількість людей | Відсоток людей |
|---|---|---|
| Немає жодних підписок | 41 | 17.3% |
| Є підписки | 196 | 82.7% |

Figure 23: Streaming subscription summary - 82.7% have at least one active subscription

## ABD Respondent Profession & Field of Activity

Segmenting professions was challenging because 89.9% of respondents were aged 18-24, meaning most were students. Some listed only "student", others listed their specialization, and those already working listed only their job title. To handle this, three separate breakdowns were created:

1. **General table** - all segments combined
2. **Profession table** - "Student" as a category plus job titles of those already working
3. **Field of activity table** - students filtered by their area of study, regardless of employment status

Of the respondents: 69 are not working, 51 did not answer, and approximately 117 indicated a profession (though some may have listed their future profession rather than a current job).

| Ваша професія/сфера діяльності | |
|---|---|
| Пусті значення | 47 |
| Специфічна відповідь | 4 |
| Студент | 46 |
| IT | 34 |
| Аналітика | 17 |
| Економіка | 18 |
| Інші професії | 14 |
| Менеджмент | 7 |
| Аудит | 8 |
| Студент, Психологія | 3 |
| Право | 4 |
| Психологія | 7 |
| Викладач | 8 |
| Студент, IT | 11 |
| Студент, Економіка | 6 |
| Студент, Право | 3 |

| Ваша професія | |
|---|---|
| Пусті значення | 47 |
| Специфічна відповідь | 4 |
| Студент | 69 |
| IT | 34 |
| Аналітика | 17 |
| Економіка | 18 |
| Інші професії | 14 |
| Менеджмент | 7 |
| Аудит | 8 |
| Право | 4 |
| Психологія | 7 |
| Викладач | 8 |

| Сфера діяльності | |
|---|---|
| Пусті значення | 47 |
| Специфічна відповідь | 4 |
| Студент | 46 |
| IT | 45 |
| Аналітика | 17 |
| Економіка | 24 |
| Інші професії | 14 |
| Менеджмент | 7 |
| Аудит | 8 |
| Право | 7 |
| Психологія | 10 |
| Викладач | 8 |

Figure 24: Three-way breakdown of respondent profession and field of activity